

TUGAS LKP P11 PRAKTIKUM PBO

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Kelas : A-S1 Sistem Informasi

VolumeBalok.java

```
p13 > VolumeBalok.java > ...
1  package p13;
2
3  import javax.swing.*;
4  import java.awt.event.ActionEvent;
5  import java.awt.event.ActionListener;
6
7  public class VolumeBalok extends JFrame {
8      private JTextField tfPanjang;
9      private JTextField tfLebar;
10     private JTextField tfTinggi;
11     private JTextField tfVolume;
12     private JButton btnHitung;
13     private JButton btnReset;
14     private JButton btnKeluar;
15
16     public VolumeBalok() {
17         setTitle("Aplikasi Penghitung Volume Balok");
18         setSize(400, 300);
19         setDefaultCloseOperation(EXIT_ON_CLOSE);
20         setLayout(null); // Menggunakan null layout seperti di praktek drag-and-drop
21
22         JLabel lblRumus = new JLabel("Rumus V = p*l*t");
23         lblRumus.setBounds(20, 10, 200, 20);
24         add(lblRumus);
25
26         JLabel lblPanjang = new JLabel("Panjang");
27         lblPanjang.setBounds(20, 40, 80, 25);
28         add(lblPanjang);
29         tfPanjang = new JTextField();
30         tfPanjang.setBounds(120, 40, 200, 25);
31         add(tfPanjang);
32
33         JLabel lblLebar = new JLabel("Lebar");
34         lblLebar.setBounds(20, 70, 80, 25);
35         add(lblLebar);
36         tfLebar = new JTextField();
37         tfLebar.setBounds(120, 70, 200, 25);
38         add(tfLebar);
39
40         JLabel lblTinggi = new JLabel("Tinggi");
41         lblTinggi.setBounds(20, 100, 80, 25);
42         add(lblTinggi);
43         tfTinggi = new JTextField();
44         tfTinggi.setBounds(120, 100, 200, 25);
45         add(tfTinggi);
46     }
```

```

47     JLabel lblVolume = new JLabel("Volume");
48     lblVolume.setBounds(20, 130, 80, 25);
49     add(lblVolume);
50     tfVolume = new JTextField();
51     tfVolume.setBounds(120, 130, 200, 25);
52     tfVolume.setEditable(false);
53     add(tfVolume);
54
55     btnReset = new JButton("Reset");
56     btnReset.setBounds(20, 170, 80, 25);
57     add(btnReset);
58
59     btnHitung = new JButton("Hitung");
60     btnHitung.setBounds(120, 170, 200, 25);
61     add(btnHitung);
62
63     btnKeluar = new JButton("Keluar");
64     btnKeluar.setBounds(20, 210, 80, 25);
65     add(btnKeluar);
66
67     // Action Listener dari modul (Langkah r-w)
68     btnHitung.addActionListener(new ActionListener() {
69         @Override
70         public void actionPerformed(ActionEvent e) {
71             // deklarasi variabel
72             double panjang, lebar, tinggi, volume;
73             // setting input
74             panjang = Double.parseDouble(tfPanjang.getText());
75             lebar = Double.parseDouble(tfLebar.getText());
76             tinggi = Double.parseDouble(tfTinggi.getText());
77             // hitung volume
78             volume = panjang * lebar * tinggi;
79             // setting output volume
80             tfVolume.setText(String.valueOf(volume));
81         }
82     });
83
84     btnReset.addActionListener(new ActionListener() {
85         @Override
86         public void actionPerformed(ActionEvent e) {
87             tfPanjang.setText("");
88             tfLebar.setText("");
89             tfTinggi.setText("");
90             tfVolume.setText("");
91             tfPanjang.requestFocus();
92         }
93     });
94
95     btnKeluar.addActionListener(new ActionListener() {
96         @Override
97         public void actionPerformed(ActionEvent e) {
98             System.exit(0);
99         }
100     });
101
102     setVisible(true);
103 }
104
105 Run | Debug
106 public static void main(String[] args) {
107     new VolumeBalok();
108 }
109

```

 Aplikasi Penghitung Volume Balok

Rumus $V = p \cdot l \cdot t$

Panjang

10

Lebar

10

Tinggi

10

Volume

1000.0

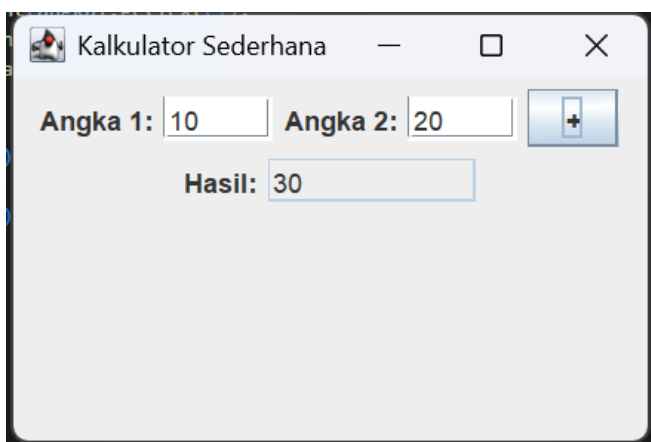
Reset

Hitung

Keluar

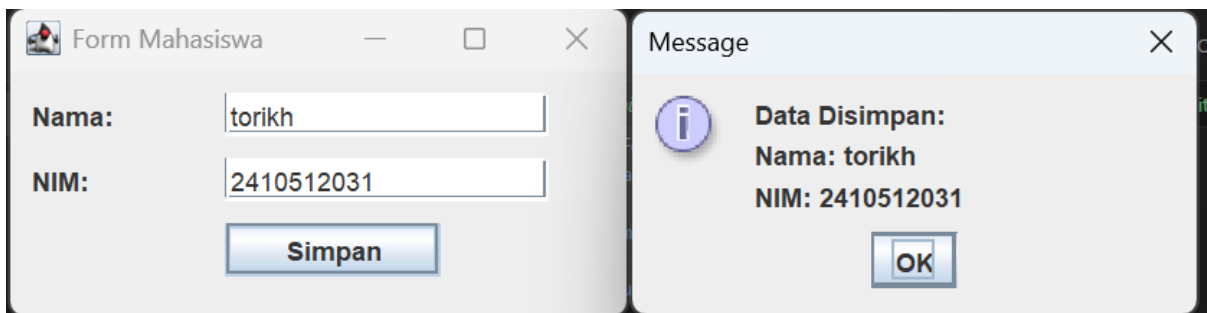
Kalkulator.java

```
p13 > Kalkulator.java > {} p13
1  package p13;
2
3  import javax.swing.*;
4  import java.awt.FlowLayout;
5
6  public class Kalkulator extends JFrame {
7      JTextField angka1, angka2, hasil;
8      JButton tambah;
9
10     public Kalkulator() {
11         setTitle("Kalkulator Sederhana");
12         setSize(300, 200);
13         setDefaultCloseOperation(EXIT_ON_CLOSE);
14         setLayout(new FlowLayout());
15
16         angka1 = new JTextField(5);
17         angka2 = new JTextField(5);
18         hasil = new JTextField(10);
19         hasil.setEditable(false);
20         tambah = new JButton("+");
21
22         tambah.addActionListener(e -> {
23             int a = Integer.parseInt(angka1.getText());
24             int b = Integer.parseInt(angka2.getText());
25             hasil.setText(String.valueOf(a + b));
26         });
27
28         add(new JLabel("Angka 1:"));
29         add(angka1);
30         add(new JLabel("Angka 2:"));
31         add(angka2);
32         add(tambah);
33         add(new JLabel("Hasil:"));
34         add(hasil);
35
36         setVisible(true);
37     }
38
39     Run | Debug
40     public static void main(String[] args) {
41         new Kalkulator();
42     }
43 }
```



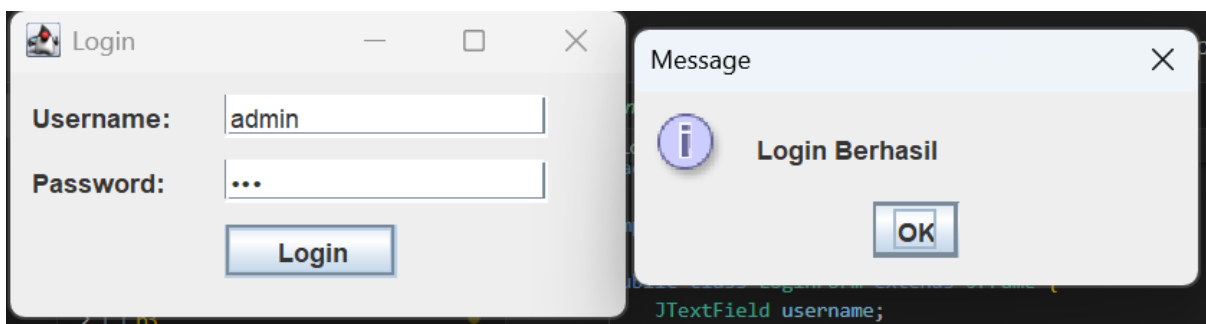
FormMahasiswa.java

```
p13 > FormMahasiswa.java > FormMahasiswa > FormMahasiswa()
1 package p13;
2
3 import javax.swing.*;
4
5 public class FormMahasiswa extends JFrame {
6     JTextField nama, nim;
7     JButton simpan;
8
9     public FormMahasiswa() {
10         setTitle("Form Mahasiswa");
11         setSize(300, 150);
12         setDefaultCloseOperation(EXIT_ON_CLOSE);
13         setLayout(null);
14
15         JLabel lblNama = new JLabel("Nama:");
16         JLabel lblNIM = new JLabel("NIM:");
17
18         nama = new JTextField();
19         nim = new JTextField();
20         simpan = new JButton("Simpan");
21
22         lblNama.setBounds(10, 10, 80, 20);
23         nama.setBounds(100, 10, 150, 20);
24         lblNIM.setBounds(10, 40, 80, 20);
25         nim.setBounds(100, 40, 150, 20);
26         simpan.setBounds(100, 70, 100, 25);
27
28         simpan.addActionListener(e ->
29             JOptionPane.showMessageDialog(this, "Data Disimpan:\nNama: " + nama.getText() + "\nNIM: " + nim.getText())
30         );
31
32         add(lblNama);
33         add(nama);
34         add(lblNIM);
35         add(nim);
36         add(simpan);
37
38         setVisible(true);
39     }
40
41     Run | Debug
42     public static void main(String[] args) {
43         new FormMahasiswa();
44     }
45 }
```



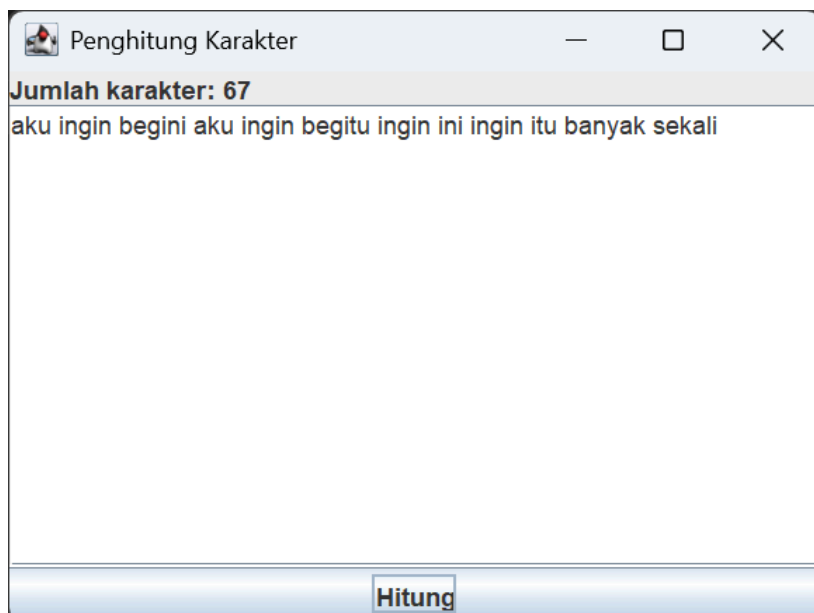
LoginForm.java

```
p13 > LoginForm.java > LoginForm > LoginForm()
1 package p13;
2
3 import javax.swing.*;
4
5 public class LoginForm extends JFrame {
6     JTextField username;
7     JPasswordField password;
8     JButton login;
9
10    public LoginForm() {
11        setTitle("Login");
12        setSize(300, 150);
13        setDefaultCloseOperation(EXIT_ON_CLOSE);
14        setLayout(null);
15
16        JLabel lblUser = new JLabel("Username:");
17        JLabel lblPass = new JLabel("Password:");
18        username = new JTextField();
19        password = new JPasswordField();
20        login = new JButton("Login");
21
22        lblUser.setBounds(10, 10, 80, 20);
23        username.setBounds(100, 10, 150, 20);
24        lblPass.setBounds(10, 40, 80, 20);
25        password.setBounds(100, 40, 150, 20);
26        login.setBounds(100, 70, 80, 25);
27
28        login.addActionListener(e -> {
29            String user = username.getText();
30            String pass = new String(password.getPassword());
31            if (user.equals("admin") && pass.equals("123")) {
32                JOptionPane.showMessageDialog(this, "Login Berhasil");
33            } else {
34                JOptionPane.showMessageDialog(this, "Login Gagal");
35            }
36        });
37
38        add(lblUser);
39        add(username);
40        add(lblPass);
41        add(password);
42        add(login);
43
44        setVisible(true);
45    }
46
47    Run | Debug
48    public static void main(String[] args) {
49        new LoginForm();
50    }
```



HitungKarakter.java

```
p13 > HitungKarakter.java > {} p13
1  package p13;
2
3  import javax.swing.*;
4  import java.awt.BorderLayout;
5
6  public class HitungKarakter extends JFrame {
7      JTextArea area;
8      JLabel hasil;
9
10     public HitungKarakter() {
11         setTitle("Penghitung Karakter");
12         setSize(400, 300);
13         setDefaultCloseOperation(EXIT_ON_CLOSE);
14         setLayout(new BorderLayout());
15
16         area = new JTextArea();
17         hasil = new JLabel("Jumlah karakter: 0");
18         JButton hitung = new JButton("Hitung");
19
20         hitung.addActionListener(e -> {
21             String text = area.getText();
22             hasil.setText("Jumlah karakter: " + text.length());
23         });
24
25         add(new JScrollPane(area), BorderLayout.CENTER);
26         add(hitung, BorderLayout.SOUTH);
27         add(hasil, BorderLayout.NORTH);
28
29         setVisible(true);
30     }
31
32     public static void main(String[] args) {
33         new HitungKarakter();
34     }
35 }
```



source code: <https://github.com/Torikh42/pbo>